TEMASEK POLYTECHNIC

SCHOOL OF INFORMATICS & IT

DIPLOMA IN APPLIED ARTICIFICAL INTELLIGENCE
DIPLOMA IN BIG DATA & ANALYTICS
DIPLOMA IN COMMON ICT PROGRAMME
DIPLOMA IN CYBER & DIGITAL FORENSICS
DIPLOMA IN IMMERSIVE MEDIA & GAME DEVELOPMENT
DIPLOMA IN INFORMATION TECHNOLOGY

AY 2025/2026 APRIL SEMESTER

Project

CYBERSECURITY FUNDAMENTALS (CCF1C03)
SUBJECT LEVEL: 1

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 7 pages (excluding cover page and content page).

2. This document describes the **Project** that you need to submit for CYFUN which contribute 60% of your overall subject assessment

3. This is an **individual assessment**; you are to complete it on your own without collaboration.

4. Please read it carefully and follow through the submission instructions.

Contents

1 PURPOSE OF THE ASSESSMENTS 1
2 KEY DATES 1
3 REQUIREMENT FOR SUCCESSFUL COMPLETION OF THIS SUBJECT 1
3.1 How do I track my progress in this subject? 1
4 PROJECT (60 MARKS) 1
4.1 Overview 1
5 Deliverables 2
5.1 Overview 2
Task A. LinkedIn Profile (5 marks) 2
Task B. Complete Windows 11: Security Lab (5 marks) 3
Task C. CIS Benchmark (20 marks) 3
Task D. Network Services (10 marks) 4
Task E. Report (10 marks) 6
Task F. Q&A (10 marks) 6
6 SCORING CRITERIA 6
7 SUBMISSION AND RETURN OF ASSESSMENT 7
7.1 Submission of Assessment 7
7.2 Late submissions 7
7.3 Academic Integrity 7

1 PURPOSE OF THE ASSESSMENTS

This project allows students to apply their skills in configuring and securing a Windows system over several weeks. By completing lab exercises, implementing key network services, and applying CIS Benchmarks, students will gain hands-on experience in system hardening. They will also complete a LinkedIn Learning certification, document their setup, and present their work through a Q&A session. The project aims to build technical confidence, encourage independent learning, and develop the ability to explain and justify technical decisions clearly.

2 KEY DATES

Please ensure that you keep all confirmation emails from TP LMS as evidence of the date/time of your submissions.

Components	Date
Case study (40%) • Immersive Lab • Poster	Due 5 Jun 2025 (Thu), 23:59, TP LMS
Project (60%) • Report • Demo	 Due 31 Jul 2025 (Thu), 23:59, TP LMS Week 16 or 17, in class

3 REQUIREMENT FOR SUCCESSFUL COMPLETION OF THIS SUBJECT

In order to pass this subject, you must:

- Attend at least 85% attendance
- Achieve an overall percentage of 50% or more for Continuous Assessments (CA)

3.1 How do I track my progress in this subject?

You are encouraged to obtain feedback from your tutor and ask questions related to the requirements of the assessments.

4 PROJECT (60 MARKS)

4.1 Overview

To be future pandemic ready, the company has mandated the employee to work from home on somedays. This work arrangement could potentially increase the cyberattacks vectors due to insecure remote access that could affect the company security posture. The current IT team are mostly new hire; thus their knowledge and skill were limited to advice or help employees to harden their laptop security. As such, the Chief Information Security Officer (CISO) wants to address these issues. The IT team should be knowledgeable and skilful with the Windows Security by completing the below Tasks A to D.

5 Deliverables

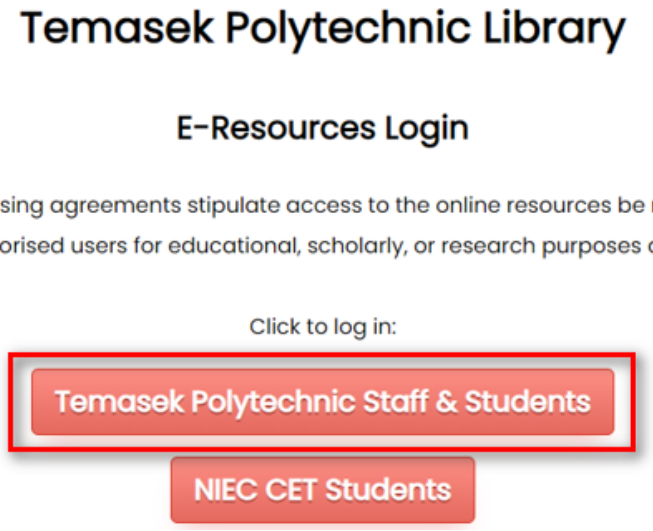
5.1 Overview

Task	Description	Remarks	Marks
A	LinkedIn Windows certification	Linked to LinkedIn and Cert of Completion	5
B	Complete week 7 & 11 Labs	To be done in VM (assessed during demo)	5
C	CIS Benchmark		20
D	Network Services		10
E	Report	Proof of Tasks done	10
F	Q&A	Assessed during Demo	10
	Total		**60 Marks**

Task A. LinkedIn Profile (5 marks)

You are to complete the following online courses and update the certificate of completion in your LinkedIn Profile.

i. Access LinkedIn Learn course via the TP Library LinkedIn Learn portal.

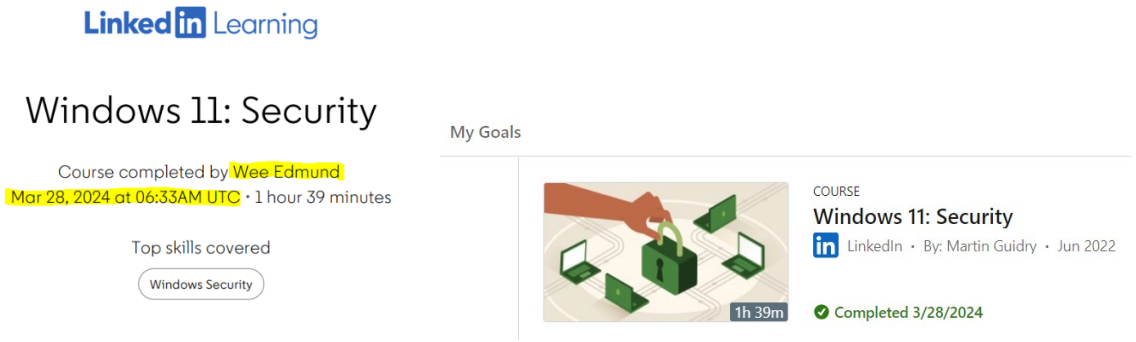


ii. Windows 11 Security course via the TP Library LinkedIn Learn portal



Upon completion of the course, you **must** be able to **show the certificate of completion** and **linked it to your LinkedIn Profile**.

Windows certification



Task B. Complete Windows 11: Security Lab (5 marks)

i. Set the computer name to CYFUN-P0x-AdminNo, e.g. CYFUN-P01-[redacted]
ii. Complete **all** the activities in Week 7 and Week 11

Task C. CIS Benchmark (20 marks)

i. In the lab VM, run the CIS tool (CIS-CAT Lite provided in LMS) using **Level 1 Corporate/Enterprise Environment** profile.
ii. Generate and save **ONE** CIS Benchmark report **before** rectification
iii. Remediate and rectify at least 2 failed items from each CIS Benchmark report categories listed below. **The overall "Fail" value for each CIS Benchmark report categories should reduce at least by 2 after remediation**. State the Fail item you have chosen to fix and explain the purpose of the remediation.

1. Account Policies
2. Local Policies
5. System Services
9. Windows Firewall with Advanced Security
17. Advance Audit Policy Configuration

iv. Generate and save **ONE** final CIS Benchmark report **after** rectifications are completed.
v. Screenshot the before and after rectification for each of the category.

vi. Following is screenshots example of the rectifications done for *Account Policies*

Before rectification

Description	Tests				Scoring		
	Pass	Fail	Error	Unkn.	Score	Max	Percent
1 Account Policies	3	4	0	2	3.0	9.0	33%

After rectification

Description	Tests				Scoring		
	Pass	Fail	Error	Unkn.	Score	Max	Percent
1 Account Policies	5	2	0	2	5.0	9.0	56%

Task D. Network Services (10 marks)

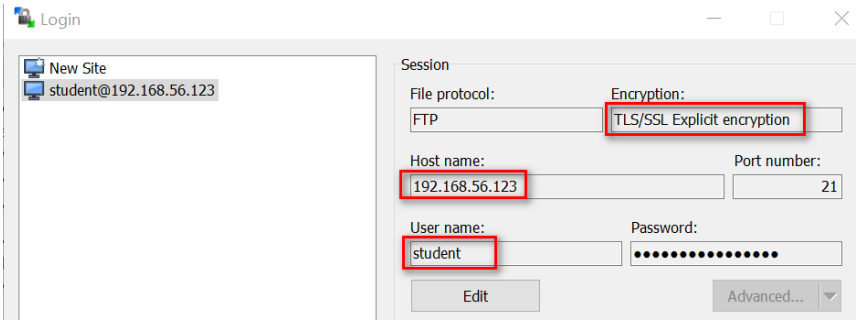
Note: This section will assess if you can research independently on what are secure FTP and web service, SSL/TLS, and Firewall rules. It will assess your capability if you can configure these services to function without guidance from your tutor. The references given are NOT working solution for your project, they are examples for you to reference and adapt to your project context.

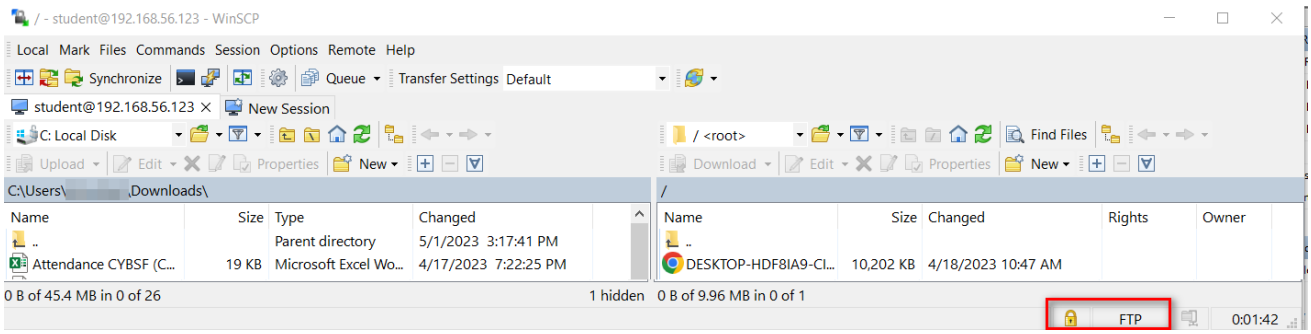
You should document the following **with screenshots:**

Demonstrate that you can:

i. **Set up secure IIS FTP service with SSL/TLS using self-signed certificate.**

- Research on what is Virtual Box Host Only Adapter
- Research on what is SSL/TLS and how it works
- IIS FTP server root directory is the VM Downloads directory
- Download and install WinSCP client on your laptop
- FTP over SSL as user "student" to the Win VM Host Only Adapter network (192.168.56.123)
- FTP uploads any file from the laptop to the Win VM successfully



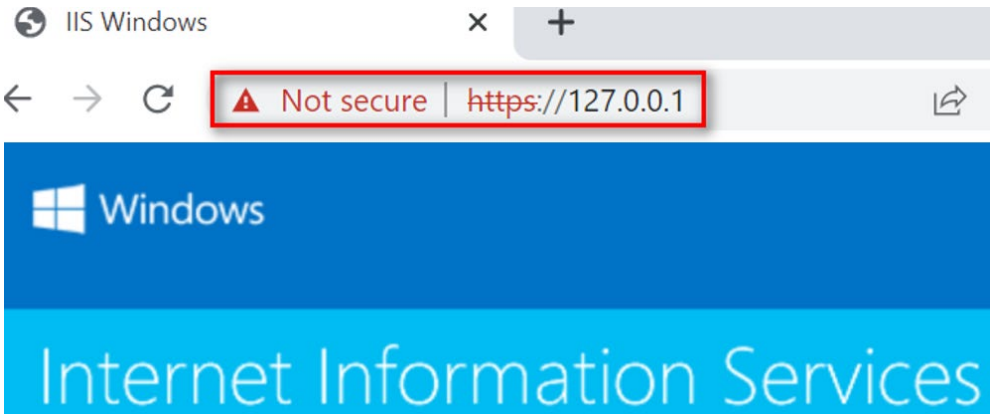


The following is for your reference:
https://winscp.net/eng/docs/guide_windows_ftps_server#on_windows_desktop_windows_11_windows_10_windows_8_windows_7_and_windows_vista

The references given are NOT working solution for your project, they are examples for you to reference and adapt to your project context.

ii. Set up secure IIS web service with SSL using self-signed certificate.

- Research on what is Network Address Translation (NAT)
- Research on what is localhost 127.0.0.1
- Research on what is Port Forwarding
- Setup IIS SSL self-signed website on your Win VM to listen on VM NAT Adapter network (10.0.2.15)
- Note that VM NAT IP (10.0.2.15) is not directly accessible by the laptop, the laptop can only access it via VBox TCP port forwarding
- Using your laptop browser, go to localhost https://127.0.0.1 to access the IIS SSL self-signed website (Demonstrate you can browse from your laptop to the Win VM IIS self-signed SSL website on the VM NAT Adapter network (10.0.2.15) via VBox localhost port forwarding using the 127.0.0.1 IP address)



The following is for your reference:

- https://www.itechguides.com/how-to-install-iis-in-windows-11/#:~:text=How%20Do%20I%20Enable%20IIS,Then%20click%20the%20OK%20button
- https://www.activecountermeasures.com/port-forwarding-with-virtualbox/

The references given are NOT working solution for your project, they are examples for you to reference and adapt to your project context.

Task E. Report (10 marks)

You are to submit the report via LMS. The file name should be **P0x_AdminNo_Name.doc** where x is your class number. You will present your documentation and demo your VM to your class tutor.

The documentation should not be more than **25** A4 size pages (**excluding** Appendix and References), font size 12, Times New Roman. Refer to the report template provided on the documentation content required.

Task F. Q&A (10 marks)

During the project demo, your tutor will be asking you questions with regards to your project. You will be assessed based on the quality of your response.

6 SCORING CRITERIA

Following are the qualities we are looking for in this project:

1. Able to show on LinkedIn Profile certificates of completion.
2. Able to configure the VM and Win OS according to project requirements.
3. Documentation with screenshots explained.
4. Well-formatted and readable report.
5. Able to articulate and answer well during Q&A.
6. Able to demonstrate good understanding of the subject content.
7. Able to perform ad hoc technical tasks and resolve technical issues.

7 SUBMISSION AND RETURN OF ASSESSMENT

7.1 Submission of Assessment

All assessments must be submitted electronically via TP LMS by 23:59 on the due date. The assessment folders are labelled per assessment item and will be available for a designated period of time to allow submission. Students should always retain a copy of all assessments submitted. Students may **not** email assessments items.

7.2 Late submissions

Late submissions will attract penalties:

- Late and < 1 day – 10% deduction from absolute marks given for the part of the work e.g. if the assignment was worth 100 marks, you were given 75 marks for the work, after the penalty you are left with 65 marks i.e. 75 – 100x10%
- Late >= 1 and < 2 days – 20% deduction from absolute marks
- Late >= 2 days – no marks will be awarded

7.3 Academic Integrity

The Safe Assign in POLITEMall will highlight similar sentences that are copied from the websites and reports submitted by your classmate as plagiarism. Do note that we treat plagiarism seriously hence disciplinary actions (E.g. failing the subject) will be taken.

- Be warned that students caught **plagiarising** will have serious consequences.
- Cite any references or visual assets used
- Do **not** use AI to generate posters or summaries **without citations**

End of paper